

Retail Data Warehouse

Pristine Star Schema Model & Technical Data Dictionary

Document Version: 1.0.0	Architecture Pattern: Star Schema (Kimball Methodology)	Target Environment: Enterprise Analytics / BI
Status: Approved / Production-Ready	Primary Grain: Individual Transaction Line Item	Documentation Generated: May 2026

1. Executive Architectural Overview

This document details the production-grade logical Entity Relationship (ER) design for the **Retail Data Warehouse Sales Model**. The data layer utilizes a **Pristine Star Schema design**, optimized specifically for fast analytical query execution, simplified aggregation paths, and native compatibility with modern Business Intelligence platforms (e.g., Power BI, Tableau, Looker).

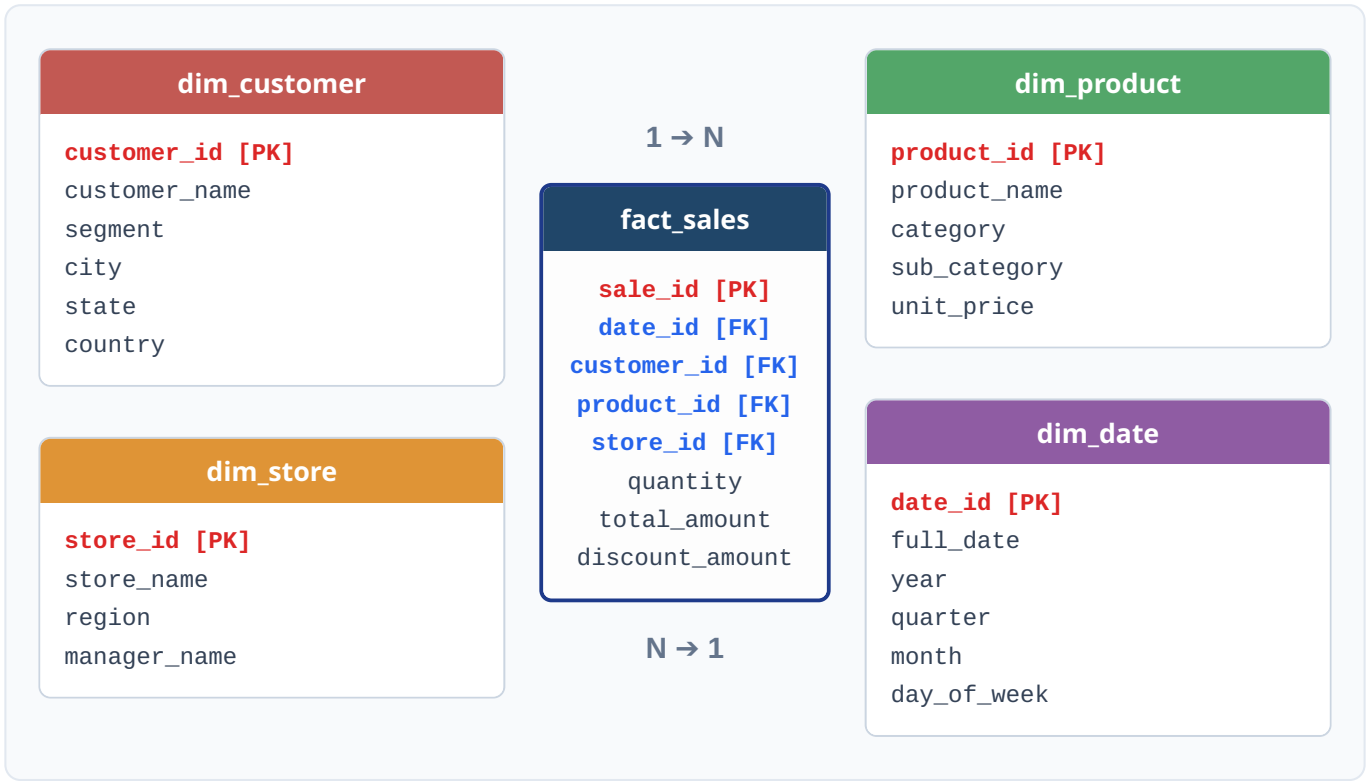
At the core of this schema is the `fact_sales` table, which records numerical business metrics captured at the atomic transactional tier. Radiating from this central table are four highly optimized dimension tables (`dim_customer`, `dim_product`, `dim_store`, and `dim_date`). This dimensional design guarantees a maximum join depth of one for any single dimension query, significantly eliminating performance overheads common in transactional normalized schemas.

Key Architectural Principles Applied:

- Surrogate Keys:** Ensured stable entity identifiers decoupled from fluctuating operational source systems.
- Strict Referential Integrity:** Enforced via 1:N declarative constraints linking dimension tables directly into the transactional fact context.
- Zoned Fields:** Explicit separation of descriptive context (Dimensions) from additive metrics (Facts).

2. Logical Schema Representation (ER Layout)

The following architectural diagram illustrates the relational layout, field distribution, key definitions, and navigational vectors established across the database schema:



3. Comprehensive Technical Data Dictionary

The following data dictionary provides an authoritative specification for every table within the model, outlining target physical data types, nullability rules, explicit key categorizations, and detailed analytical functions.

3.1. Central Table: fact_sales

Description: Tracks granular line-item checkout records capturing transactional volume and baseline financial realizations.

Column Name	Data Type	Key Class	Nullability	Functional Definition & Business Context
sale_id	BIGINT	PK	NOT NULL	Surrogate unique identifier generating tracking sequences for each line transaction.
date_id	INT	FK	NOT NULL	References calendar time components managed natively inside dim_date.
customer_id	INT	FK	NOT NULL	Maps unique consumer properties associated with order actions back to dim_customer.
product_id	INT	FK	NOT NULL	Identifies target inventory items sold; validates entity against dim_product.
store_id	INT	FK	NOT NULL	Locates the point-of-sale brick-and-mortar operations originating records from dim_store.
quantity	INT	None	NOT NULL	Additive numerical count representing the precise product unit volumes checked out.
total_amount	DECIMAL(18,2)	None	NOT NULL	Gross absolute monetary value realizations computed before discounts apply.
discount_amount	DECIMAL(18,2)	None	NOT NULL	Deductive markdown concessions subtracted directly from line transactions.

3.2. Dimension Table: dim_customer

Description: Maintains long-lived descriptive attributes mapping customer demography and market segmentation profiles.

Column Name	Data Type	Key Class	Nullability	Functional Definition & Business Context
customer_id	INT	PK	NOT NULL	Unique surrogate identifier generated to decouple core warehouse tracking from upstream CRM IDs.

Column Name	Data Type	Key Class	Nullability	Functional Definition & Business Context
customer_name	VARCHAR(150)	None	NOT NULL	Full client name or legal corporate entity name string.
segment	VARCHAR(50)	None	NOT NULL	Strategic classification sorting buyers into distinct behaviors (e.g., Retail, B2B, VIP).
city	VARCHAR(100)	None	NOT NULL	Primary address city designation recorded for the client entity.
state	VARCHAR(100)	None	NOT NULL	Provincial or territory administrative boundary marking user residency.
country	VARCHAR(100)	None	NOT NULL	Highest level geographical taxonomy code for localized reporting.

3.3. Dimension Table: dim_product

Description: Contains product listing specifications and categorization levels allowing item analysis hierarchies.

Column Name	Data Type	Key Class	Nullability	Functional Definition & Business Context
product_id	INT	PK	NOT NULL	Primary internal warehouse identification key assigned to structural stock units (SKUs).
product_name	VARCHAR(255)	None	NOT NULL	Full descriptive title of inventory merchandise item.
category	VARCHAR(100)	None	NOT NULL	High-level structural segment grouping (e.g., Apparel, Electronics, Groceries).
sub_category	VARCHAR(100)	None	NOT NULL	Detailed item subset category allowing deep portfolio drill-down analysis.
unit_price	DECIMAL(14,2)	None	NOT NULL	Base standardized currency value assigned per product asset unit.

3.4. Dimension Table: dim_store

Description: Identifies details and operations hierarchies characterizing physical storefront locations.

Column Name	Data Type	Key Class	Nullability	Functional Definition & Business Context
store_id	INT	PK	NOT NULL	Surrogate identifying master sequence assigned to uniquely map active physical business outlets.
store_name	VARCHAR(150)	None	NOT NULL	Human-readable operational naming identifier specific to the storefront branch.
region	VARCHAR(50)	None	NOT NULL	Corporate operating region mapping boundaries (e.g., North-East, APAC, EMEA).
manager_name	VARCHAR(150)	None	NOT NULL	Full name metadata referencing the general manager responsible for site operations.

3.5. Dimension Table: dim_date

Description: Time-intelligence framework providing calendar transformations decoupled from SQL computational functions.

Column Name	Data Type	Key Class	Nullability	Functional Definition & Business Context
date_id	INT	PK	NOT NULL	Smart surrogate key integer mapped via chronological format patterns (YYYYMMDD).
full_date	DATE	None	NOT NULL	Standard foundational SQL structure timestamp date indicator.
year	INT	None	NOT NULL	Four-digit calendar calculation year integer (e.g., 2026).
quarter	VARCHAR(2)	None	NOT NULL	Calendar quarter categorization code tracking periods (e.g., Q1, Q2, Q3, Q4).
month	INT	None	NOT NULL	Numerical position mapping sequence assigned to annual months (range: 1 - 12).
day_of_week	VARCHAR(15)	None	NOT NULL	Literal context name describing active day parameters (e.g., Monday, Sunday).

4. Relationship Matrix & Integrity Specifications

To preserve structural accuracy and prevent orphan records from polluting aggregates, the schema enforces declarative referential integrity. All analytical transactions must conform to the 1-to-Many configurations mapped below:

Parent Dimension Table	Primary Join Key	Child Fact Table Key Reference	Enforced Cardinality	On-Update / Delete Strategy
dim_customer	customer_id	fact_sales.customer_id	1 : Many (1:N)	RESTRICT / NO ACTION
dim_product	product_id	fact_sales.product_id	1 : Many (1:N)	RESTRICT / NO ACTION
dim_store	store_id	fact_sales.store_id	1 : Many (1:N)	RESTRICT / NO ACTION
dim_date	date_id	fact_sales.date_id	1 : Many (1:N)	RESTRICT / NO ACTION

5. Calculated Key Performance Indicators (KPIs) & Analytics Value

The flat layout of this clean model permits rapid aggregation formulations using standard SQL functions or specialized analytics layer calculations (such as DAX expressions or MDX scripts). Primary commercial indicators are computed using the formulas specified below:

1. Net Realized Revenue

```
SUM(fact_sales.total_amount) - SUM(fact_sales.discount_amount)
```

2. Average Unit Pricing (AUP) Realization

```
SUM(fact_sales.total_amount) / SUM(fact_sales.quantity)
```

3. Promo Markdown Blended Discount Rate

```
SUM(fact_sales.discount_amount) / SUM(fact_sales.total_amount)
```